

CLEMENT HERITIER

Embedded Systems & Robotics Engineer

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Paris, FR

EXPERIENCE

VIE – Lab Test Automation Engineer

Alstom

June 2025 – November 2026

Pittsburgh, US

- Designed and developed a full-stack network analysis platform, enabling real-time packet capture, visualization, and diagnostics for complex embedded communication systems.
- Automated end-to-end validation workflows in virtualized environments, integrating Robot Framework to improve test scalability, repeatability, and regression coverage.
- Built API-driven backend services and intuitive GUI interfaces, significantly reducing debug time and improving traceability of system-level issues.
- Collaborated with cross-functional teams (software, systems, validation) to identify communication issues and enhance overall system reliability.

Internship – Model Based & Function Development Engineer

Bosch

September 2024 – March 2025

Paris, FR

- Supported deployment of model-based design workflows (MATLAB/Simulink) across distributed teams, ensuring compliance with automotive development standards.
- Improved model test coverage (MCDRC) using Simulink Design Verifier, contributing to higher software safety and validation robustness.
- Diagnosed and resolved execution and compilation issues during Processor-in-the-Loop (PIL) testing on embedded targets.

Internship – Embedded Systems Engineer

Andrice

May 2023 – August 2023

Lyon, FR

- Designed a 500 W power converter and developed PCB (Altium).
- Led prototype bring-up, testing, and system validation.
- Performed electrical and thermal testing to ensure reliability.

PROJECTS

Autonomous Robotic Waste Collection System

May 2024 – August 2024

Heriot-Watt University | Edinburgh, UK

- Developed navigation and control algorithms for an autonomous surface vehicle.
- Integrated embedded software with sensors and actuators.
- Validated system through real-world testing and performance analysis.

“Voix de l’Univers”

April 2022 – June 2022

ESIEE | Paris, FR

- Designed an embedded system to detect electromagnetic signals.
- Developed circuits, PCB, and performed simulations.
- Acquired and analyzed real-world signal data.

EDUCATION

Master degree in Robotics

Heriot-Watt University

Sep 2023 – Sep 2024

Edinburgh, UK

Master degree in Embedded & Electrical Systems

ESIEE Paris

Sep 2021 – Sep 2024

Paris, FR

Preparatory Classes in Mathematics & Physics (CPGE)

Camille Vernet

Sep 2018 – Sep 2021

Valence, FR

Scientific Baccalauréat – Honors

Institut Notre-Dame

Sep 2018

Valence, FR

SKILLS

Programming

Python C/C++ HTML/CSS Java VHDL

Embedded & Electronics

STM32 ESP32/ESP8266 Altium Designer LTSpice

Software & Tools

MATLAB/Simulink ROS Robot Framework Git/GitHub Docker IBM Rhapsody

Languages

French (Native) English (Professional) Spanish (Basic)

CERTIFICATIONS

- TOEIC (2023) — Score: 905
- Voltaire (French) Certification

INTERESTS

- Competitive Rugby (15 years)
- Robotics & Aerospace
- International Travel